Sulabham · APPSC AEE 2026 — Paper 2 (Civil & Mechanical Common)

Module 1 Test: Strength of Materials (2023 exam trend)

Based on the official APPSC AEE Paper II held on 22 Aug 2023 (Shift 1) · take this test at the end of Week 3

Part 1: How APPSC asked Strength of Materials in 2023

In the 2023 Paper II, **51 of the 150 questions (one-third of the paper)** came from Strength of Materials (SOM) — the single biggest block. The rest was Fluid Mechanics & Hydraulic Machines. Topic-wise count:

Topic	Qs in 2023	Priority for 2026
Simple stresses & strains, elastic constants, stress–strain curve	14	**Very high**
Deflection of beams & structural analysis (double integration, moment-area, conjugate beam, Castigliano)	11	**Very high** — not named in Paper 2 syllabus, but heavily asked
Shear force & bending moment	6	High
Principal stresses & Mohr's circle	5	High
Bending stresses (section modulus, $\sigma = Ey/R$)	4	High
Theories of failure	4	High
Strain energy (gradual / sudden load, resilience)	4	High
Columns (Euler)	2	Medium
Torsion	1	Medium — under-asked in 2023, may increase
Thin cylinders, springs, lifting machines	0	Medium — in syllabus, keep formulas ready

How the questions were framed (types seen in 2023):

Question type	Approx. share	Real 2023 example (shortened)
Direct numerical	30%	"A steel plate 120 mm × 20 mm is bent into a circular arc of radius 10 m. Find max bending stress ($E = 2\times10^5$)." → 200 N/mm²
Formula identification	20%	"Crippling load for a column with both ends fixed?" → $4\pi^2EI/l^2$
Definition / concept	25%	"Stress causing a permanent extension of 0.1–0.2% of gauge length?" → Proof stress
Which is INCORRECT / wrongly matched	10%	"Assumption of Euler's theory that is INCORRECT?" → direct stress is very large compared with bending stress
Statement 1 / Statement 2 (true or false)	8%	"S1: Max principal stress theory for brittle materials; S2: Max shear stress theory for ductile…" → all true
Figure-based	7%	Identify point P on a mild steel stress–strain curve; pick the correct BMD for an SS beam with UDL

Five lessons from the 2023 paper

1. **No calculator is allowed.** The numbers are chosen to cancel neatly (e.g., $\pi = 3.14$ with a load of 31.4 kN). Practise doing the arithmetic by hand.
2. **Many numericals are standard textbook solved examples.** Solve every worked example in R.K. Bansal (Strength of Materials) — chapters on stress, strain energy, SF/BM, bending, deflection.
3. **Formula recall decides a large share of marks.** Questions like 'deflection of a fixed beam with central load' come directly. Memorise the formula boxes from Module 1 and the deflection box below.
4. **Learn the 'reverse' statements.** INCORRECT-type questions take a true assumption and flip one word (small ↔ large, least ↔ maximum, twice ↔ half).
5. **Negative marking is 1/3.** In 2023 the papers were mostly straightforward, so accuracy matters more than attempting all 150.

Must-know deflection formulas (not in Module 1 — learn these now)

Cantilever, point load W at free end: $\delta = WL^3/3EI$, slope $\theta = WL^2/2EI$ **Cantilever, UDL w:** $\delta = wL^4/8EI$, $\theta = wL^3/6EI$

SS beam, central point load W: $\delta = WL^3/48EI$, $\theta = WL^2/16EI$ **SS beam, UDL w:** $\delta = 5wL^4/384EI$, $\theta = wL^3/24EI$

Fixed beam, central point load: $\delta = WL^3/192EI$ **Fixed beam, UDL:** $\delta = wL^4/384EI$ (fixed beam deflects 1/4 and 1/5 of SS beam)

Differential equation of elastic curve: $EI \cdot d^2y/dx^2 = M$ (assumes Hooke's law, small deflection, shear deformation neglected)

Methods: double integration, Macaulay's method, moment-area (Mohr's theorems), conjugate beam, strain energy (Castigliano: $\delta = \partial U/\partial W$)

Conjugate beam: fixed ↔ free, simple support stays simple; SF of conjugate beam = slope of real beam; BM of conjugate beam = deflection of real beam; same length

Statically determinate: cantilever, SS beam, overhanging beam, three-hinged arch. **Indeterminate:** fixed beam, propped cantilever, continuous beam, two-hinged & fixed arches

Part 2: Module 1 Test — 50 Questions

Time: 50 minutes (exam pace = 1 minute per question) **Marks:** +1 correct, −0.33 wrong, 0 unattempted **Calculator:** not allowed

Write your answers on a separate sheet. Don't look at the key until the timer stops. Mark a question only if you can eliminate at least two options.

A. Simple Stresses, Strains and Elastic Constants

1. A steel bar 2 m long has a rectangular cross-section 20 mm × 25 mm. It is subjected to an axial pull of 50 kN. Taking $E = 2 \times 10^5$ N/mm², the elongation of the bar is:

(a) 0.5 mm (b) 1.0 mm (c) 1.5 mm (d) 2.0 mm

2. A material has modulus of elasticity E = 200 GPa and modulus of rigidity G = 80 GPa. Its Poisson's ratio is:

(a) 0.30 (b) 0.33 (c) 0.25 (d) 0.20

3. For a material with E = 200 GPa and Poisson's ratio 0.25, the bulk modulus K is:

(a) 100 GPa (b) 160 GPa (c) 266.7 GPa (d) 133.3 GPa

4. Consider the following statements regarding Poisson's ratio (μ):

I. For most metals, μ lies between 0.25 and 0.33.

II. For a perfectly incompressible material, μ = 0.5.

III. For an isotropic elastic material, μ can be greater than 0.5.

IV. For cork, μ is nearly zero.

Which of the above statements are correct?

(a) I, II and IV only (b) I and II only (c) I, III and IV only (d) I, II, III and IV

5. The strain energy stored per unit volume of a body when it is stressed up to the elastic limit is called:

(a) resilience (b) proof resilience

(c) modulus of resilience (d) modulus of toughness

6. A short RCC column has a concrete area of 80,000 mm² and steel area of 2,000 mm². The modular ratio is 10 and the column carries an axial load of 500 kN. The stress in steel is:

(a) 5 N/mm² (b) 25 N/mm² (c) 50 N/mm² (d) 100 N/mm²

7. A steel rail 10 m long is laid at 20°C with its ends fixed. The temperature rises to 60°C, but the supports yield by 0.4 mm. Take $\alpha = 12 \times 10^{-6}$ /°C and $E = 2 \times 10^5$ N/mm². The stress in the rail is:

(a) 96 N/mm² (b) 88 N/mm² (c) 80 N/mm² (d) 8 N/mm²

8. A uniformly tapering circular rod 1 m long has diameters 40 mm and 20 mm at its ends. It carries an axial pull of 31.4 kN. Taking $E = 2 \times 10^5$ N/mm² and π = 3.14, its elongation is:

(a) 0.50 mm (b) 1.0 mm (c) 0.125 mm (d) 0.25 mm

9. A uniform bar of length L hangs vertically from a support. The elongation due to its own weight W is _____ the elongation that would occur if the same load W were applied at its lower end.

(a) equal to (b) twice (c) half of (d) one-fourth of

10. Match List-I (property) with List-II (meaning):

List-I	List-II
A. Ductility	1. Energy absorbed up to fracture
B. Malleability	2. Can be drawn into thin wires
C. Brittleness	3. Can be hammered into thin sheets
D. Toughness	4. Fractures with very little deformation

Codes:

(a) A-2, B-3, C-4, D-1 (b) A-3, B-2, C-4, D-1 (c) A-2, B-3, C-1, D-4 (d) A-3, B-2, C-1, D-4

B. Strain Energy and Types of Loading

11. An axial load of 50 kN is applied suddenly to a bar of cross-sectional area 1000 mm². The maximum stress induced is:
(a) 50 N/mm² (b) 100 N/mm² (c) 150 N/mm² (d) 200 N/mm²

12. A load of 10 kN falls through a height of 2 mm onto a collar attached to a vertical bar 1 m long with area 1000 mm². Take $E = 2 \times 10^5$ N/mm². The maximum impact stress in the bar is:
(a) 10 N/mm² (b) 20 N/mm² (c) 90 N/mm² (d) 100 N/mm²

13. A bar of cross-sectional area 500 mm² and length 2 m is stressed to 100 N/mm². With $E = 2 \times 10^5$ N/mm², the strain energy stored in the bar is:
(a) 12.5 N·m (b) 50 N·m (c) 100 N·m (d) 25 N·m

14. Consider the following statements:
I. For the same load, the stress produced by a suddenly applied load is twice, and the strain energy stored is four times, that produced by a gradually applied load.
II. Proof resilience is the maximum strain energy that can be stored in a body up to the elastic limit.
III. Modulus of resilience is a dimensionless quantity.
IV. The impact stress formula reduces to the sudden-load stress when the height of fall is zero.
Which of the above statements are correct?
(a) I and II only (b) I, II and IV only (c) II, III and IV only (d) I, II, III and IV

15. Assertion (A): Mild steel is preferred over cast iron for members subjected to shock loads.
Reason (R): Cast iron has a higher modulus of elasticity than mild steel.
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is NOT the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

C. Shear Force and Bending Moment

16. A simply supported beam of span 6 m carries a UDL of 20 kN/m over the entire span. The maximum bending moment is:
(a) 180 kN·m (b) 120 kN·m (c) 90 kN·m (d) 45 kN·m

17. A cantilever 3 m long carries a point load of 10 kN at its free end and a UDL of 4 kN/m over its whole length. The bending moment at the fixed end is:
(a) 48 kN·m (b) 30 kN·m (c) 39 kN·m (d) 66 kN·m

18. A simply supported beam of span 5 m carries a point load of 30 kN at 2 m from the left support. The maximum bending moment is:
(a) 30 kN·m (b) 37.5 kN·m (c) 45 kN·m (d) 36 kN·m

19. In which of the following beams does a point of contraflexure occur?
I. Cantilever carrying UDL over its full length
II. Simply supported beam with a central point load
III. Beam overhanging on both sides, with UDL over the full length
IV. Fixed beam carrying UDL over the full span
(a) I and II only (b) III and IV only (c) II and III only (d) I, III and IV only

20. A simply supported beam AB of span 6 m carries a load varying uniformly from zero at A to 12 kN/m at B. The reaction at B is:
(a) 12 kN (b) 18 kN (c) 24 kN (d) 36 kN

21. Consider the following statements regarding a beam:
I. The slope of the shear force diagram at a section equals the intensity of loading at that section.
II. The slope of the bending moment diagram at a section equals the shear force at that section.
III. Bending moment is maximum or minimum where the shear force changes sign.
IV. Over a portion carrying no load, the bending moment diagram is parabolic.
Which of the above statements are correct?
(a) I, II and III only (b) I and II only (c) II, III and IV only (d) I, II, III and IV

22. Match List-I (loading) with List-II (shape of BMD):

List-I	List-II
A. SS beam, central point load	1. Rectangle
B. SS beam, UDL on full span	2. Triangle
C. Cantilever, UVL (zero at free end)	3. Cubic curve
D. Cantilever, couple at free end	4. Parabola

Codes:
(a) A-2, B-3, C-4, D-1 (b) A-4, B-2, C-3, D-1 (c) A-2, B-4, C-1, D-3 (d) A-2, B-4, C-3, D-1

23. A simply supported beam of span 8 m carries a UDL of 10 kN/m over its full span. The shear force at a section 2 m from the left support is:
(a) 0 (b) 20 kN (c) 40 kN (d) 60 kN

D. Bending Stresses and Shear Stresses in Beams

24. A steel plate 12 mm thick is bent into a circular arc of radius 8 m. Taking $E = 2 \times 10^5$ N/mm², the maximum bending stress induced is:

(a) 300 N/mm² (b) 200 N/mm² (c) 150 N/mm² (d) 75 N/mm²

25. A rectangular beam 100 mm wide and 200 mm deep is subjected to a maximum bending moment of 20 kN·m. The maximum bending stress is:

(a) 30 N/mm² (b) 15 N/mm² (c) 60 N/mm² (d) 45 N/mm²

26. A rectangular beam has depth twice its width. The ratio of its moment-carrying capacity when placed with the depth vertical to that when placed with the width vertical is:

(a) 1 (b) 2 (c) 4 (d) 8

27. Statement 1: For the same cross-sectional area, a square section is stronger in bending than a circular section.
Statement 2: An I-section is the most economical beam section because most of its material is placed far from the neutral axis.

(a) Both statements are true (b) Both statements are false
(c) Statement 1 is true, Statement 2 is false (d) Statement 1 is false, Statement 2 is true

28. Which of the following statements is INCORRECT?

(a) The neutral axis passes through the centroid of the section (homogeneous beam, elastic bending)
(b) Bending stress is zero at the neutral axis
(c) Shear stress is zero at the neutral axis of a rectangular section
(d) Fibres lying on the neutral layer do not change in length

29. A rectangular beam 100 mm wide and 200 mm deep carries a shear force of 40 kN. The maximum shear stress is:

(a) 2.0 N/mm² (b) 2.67 N/mm² (c) 4.0 N/mm² (d) 3.0 N/mm²

30. Consider the following statements for an I-section beam:

I. Most of the bending moment is resisted by the flanges.
II. Most of the shear force is resisted by the web.
III. The shear stress distribution shows a sudden jump at the junction of flange and web.
IV. Maximum shear stress occurs at the neutral axis.
Which of the above statements are correct?

(a) I and II only (b) I, II and III only (c) II and IV only (d) I, II, III and IV

E. Principal Stresses, Mohr's Circle and Theories of Failure

31. At a point in a strained material, $\sigma_x = 80$ N/mm² (tensile), $\sigma_y = 20$ N/mm² (tensile) and shear stress $\tau_{xy} = 40$ N/mm². The principal stresses are:

(a) 100 and 0 N/mm² (b) 90 and 10 N/mm² (c) 80 and 20 N/mm² (d) 110 and −10 N/mm²

32. A bar is subjected to a direct tensile stress of 100 N/mm². The normal stress on a plane inclined at 30° to the normal cross-section is:

(a) 25 N/mm² (b) 50 N/mm² (c) 75 N/mm² (d) 86.6 N/mm²

33. An element of a shaft is in a state of pure shear with $\tau = 60$ N/mm². The principal stresses and the planes on which they act are:

(a) ±60 N/mm², on planes at 45° to the shear planes
(b) ±30 N/mm², on planes at 45°
(c) 60 and 0 N/mm², on planes at 90°
(d) ±120 N/mm², on planes at 45°

34. Consider the following statements about Mohr's circle of stress:

I. The radius of the circle equals the maximum shear stress.
II. The centre lies on the σ-axis at $(\sigma_x + \sigma_y)/2$.
III. When $\sigma_x = \sigma_y$ and $\tau = 0$, the circle reduces to a point.
IV. An angle measured on Mohr's circle is half the actual angle between the planes.
Which of the above statements are correct?

(a) I and II only (b) I, II and III only (c) I, III and IV only (d) I, II, III and IV

35. Match List-I (theory of failure) with List-II (criterion):

List-I	List-II
A. Rankine	1. Distortion (shear strain) energy
B. Guest / Tresca	2. Maximum principal stress
C. St. Venant	3. Maximum shear stress
D. Von Mises / Hencky	4. Maximum principal strain

Codes:

(a) A-2, B-3, C-1, D-4 (b) A-2, B-3, C-4, D-1 (c) A-3, B-2, C-4, D-1 (d) A-4, B-3, C-2, D-1

36. The principal stresses at a point in an elastic material are 180 N/mm² (tensile), 60 N/mm² (tensile) and 30 N/mm² (compressive). The elastic limit in simple tension is 200 N/mm². According to the maximum shear stress theory:

(a) τ_{max} = 60 N/mm², the material is safe (b) τ_{max} = 105 N/mm², failure will occur

(c) τ_{max} = 90 N/mm², the material is safe (d) τ_{max} = 105 N/mm², the material is safe

37. Assertion (A): The maximum principal stress theory is not suitable for ductile materials.

Reason (R): Ductile materials generally fail by yielding due to shear rather than by direct tension.

(a) Both A and R are true, and R is the correct explanation of A

(b) Both A and R are true, but R is NOT the correct explanation of A

(c) A is true, but R is false

(d) A is false, but R is true

38. A plate is subjected to tensile stresses σ_x = 100 N/mm² and σ_y = 50 N/mm². Taking E = 2×10^5 N/mm² and μ = 0.3, the strain in the x-direction is:

(a) 5.0×10^{-4} (b) 4.25×10^{-4} (c) 4.5×10^{-4} (d) 3.5×10^{-4}

F. Deflection of Beams (asked heavily in 2023)

39. A cantilever 2 m long carries a point load of 20 kN at its free end. The flexural rigidity EI = 4000 kN·m². The deflection at the free end is:

(a) 6.67 mm (b) 10 mm (c) 20 mm (d) 13.33 mm

40. If the span of a simply supported beam carrying a UDL (same intensity w) is doubled, its maximum deflection becomes:

(a) 2 times (b) 4 times (c) 8 times (d) 16 times

41. Match List-I (beam and load) with List-II (maximum deflection):

List-I	List-II
A. SS beam, central point load W	1. $wL^4/8EI$
B. SS beam, UDL w	2. $WL^3/192EI$
C. Cantilever, UDL w	3. $WL^3/48EI$
D. Fixed beam, central point load W	4. $5wL^4/384EI$

Codes:

(a) A-3, B-4, C-1, D-2 (b) A-4, B-3, C-1, D-2 (c) A-3, B-4, C-2, D-1 (d) A-2, B-4, C-1, D-3

42. Consider the following statements regarding the conjugate beam method:

I. A fixed end of the real beam becomes a free end in the conjugate beam.

II. A free end of the real beam becomes a fixed end in the conjugate beam.

III. A simple end support remains a simple support.

IV. The shear force at a section of the conjugate beam gives the deflection of the real beam.

Which of the above statements are correct?

(a) I and II only (b) I, II and III only (c) III and IV only (d) I, II, III and IV

G. Columns and Struts

43. The Euler crippling load of a column with both ends hinged is 400 kN. If the same column is fixed at one end and free at the other, its crippling load is:

(a) 100 kN (b) 200 kN (c) 800 kN (d) 1600 kN

44. A solid circular column of diameter 100 mm and length 3 m is hinged at both ends. Its slenderness ratio is:

(a) 30 (b) 60 (c) 120 (d) 240

45. Which of the following statements is INCORRECT?

(a) Euler's formula is valid only for long columns

(b) Rankine's formula can be used for both short and long columns

(c) Rankine's constant depends on the material of the column

(d) Euler's crippling load is calculated using the maximum moment of inertia of the section

H. Torsion and Springs

46. A shaft rotating at 300 rpm transmits a torque of 1000 N·m. Taking π = 3.14, the power transmitted is:

(a) 15.7 kW (b) 31.4 kW (c) 62.8 kW (d) 94.2 kW

47. Consider the following statements regarding circular shafts under torsion:

I. For the same weight, a hollow shaft transmits more torque than a solid shaft.

II. The shear stress at the centre of a solid shaft is zero.

III. The shear stress varies linearly from the centre to the outer surface.

IV. The polar section modulus of a solid shaft of diameter d is $\pi d^3/32$.

Which of the above statements are correct?

(a) I, II, III and IV (b) II and III only (c) I, II and III only (d) I and IV only

48. Two close-coiled helical springs of stiffness 20 N/mm and 30 N/mm are connected in series. Under an axial load of 120 N, the total deflection is:

(a) 2.4 mm (b) 4 mm (c) 6 mm (d) 10 mm

I. Thin Cylinders and Lifting Machines

49. A thin cylindrical shell of 1 m diameter and 10 mm wall thickness is subjected to an internal pressure of 2 N/mm². The circumferential (hoop) stress is:

(a) 25 N/mm² (b) 50 N/mm² (c) 100 N/mm² (d) 200 N/mm²

50. In a screw jack with a velocity ratio of 40, an effort of 125 N lifts a load of 2000 N. The efficiency of the jack and its nature are:

(a) 40%, self-locking (b) 40%, reversible (c) 62.5%, reversible (d) 62.5%, self-locking

Part 3: Answer Key with Short Solutions

A. Simple Stresses, Strains and Elastic Constants

1. (b) $\delta = PL/AE = (50{,}000 \times 2000)/(500 \times 2\times10^5) = 10^8/10^8 = 1$ mm.

2. (c) $E = 2G(1+\mu) \rightarrow 200 = 160(1+\mu) \rightarrow \mu = 0.25$.

3. (d) $E = 3K(1-2\mu) \rightarrow K = 200/(3 \times 0.5) = 133.3$ GPa.

4. (a) For isotropic materials μ ranges 0 to 0.5, so III is wrong. Cork ≈ 0 (used as bottle stopper — it doesn't bulge).

5. (c) Proof resilience = total energy up to elastic limit; per unit volume = modulus of resilience = $\sigma^2/2E$ (2023 asked its formula).

6. (c) Equivalent area = Ac + m·As = 80,000 + 20,000 = 1,00,000 mm². $\sigma c = 5{,}00{,}000/1{,}00{,}000 = 5$ N/mm². $\sigma s = m\cdot\sigma c = 50$ N/mm².

7. (b) Free expansion = $\alpha L\Delta T = 12\times10^{-6} \times 10{,}000 \times 40 = 4.8$ mm. Prevented = 4.8 − 0.4 = 4.4 mm. $\sigma = E \times 4.4/10{,}000 = 88$ N/mm². (Trap: 96 ignores the yielding.)

8. (d) $\delta = 4PL/(\pi E\cdot d_1\cdot d_2) = (4 \times 31{,}400 \times 1000)/(3.14 \times 2\times10^5 \times 40 \times 20) = 0.25$ mm. (2023 asked this formula directly.)

9. (c) Self-weight: $\delta = WL/2AE$; end load: $\delta = WL/AE$. Ratio = ½.

10. (a) Ductility = wires; malleability = sheets; brittleness = sudden fracture (cast iron, glass); toughness = area under the full stress–strain curve.

B. Strain Energy and Types of Loading

11. (b) Sudden load: $\sigma = 2P/A = 2 \times 50{,}000/1000 = 100$ N/mm².

12. (d) $\sigma = (P/A)[1 + \sqrt{(1 + 2AEh/PL)}]$. $2AEh/PL = (2 \times 1000 \times 2\times10^5 \times 2)/(10{,}000 \times 1000) = 80$. $\sqrt{81} = 9 \rightarrow \sigma = 10 \times (1 + 9) = 100$ N/mm².

13. (d) $U = (\sigma^2/2E) \times V = (10{,}000/4\times10^5) \times (500 \times 2000) = 0.025 \times 10^6 = 25{,}000$ N·mm = 25 N·m.

14. (b) $U \propto \sigma^2$, so 2× stress → 4× energy (I true). Modulus of resilience = energy/volume = N·mm/mm³ = N/mm² — it has units of stress, so III is false. With h = 0, $\sigma = (P/A)(1+1) = 2P/A$ (IV true).

15. (c) A is true — mild steel is ductile and tough, absorbs shock. R is false — E of cast iron (≈100 GPa) is lower than steel (≈200 GPa).

C. Shear Force and Bending Moment

16. (c) $M_{max} = wL^2/8 = 20 \times 36/8 = 90$ kN·m at mid-span.

17. (a) $M = WL + wL^2/2 = 10 \times 3 + 4 \times 9/2 = 30 + 18 = 48$ kN·m (hogging).

18. (d) $M_{max} = Wab/L = 30 \times 2 \times 3/5 = 36$ kN·m, under the load. (37.5 = WL/4 trap — only for a central load.)

19. (b) Contraflexure needs BM to change sign (sagging ↔ hogging). Cantilever = only hogging; SS beam = only sagging. Overhanging and fixed beams have both.

20. (c) Total load = ½ × 12 × 6 = 36 kN acting at 2L/3 = 4 m from A. RB = 36 × 4/6 = 24 kN; RA = 12 kN (= wL/6, asked in 2023).

21. (a) $dV/dx = -w$, $dM/dx = V$, so I, II, III are true. With no load, SF is constant and BM is linear (straight line), not parabolic.

22. (d) Degree of BMD = degree of load + 2: point load → linear (triangle); UDL → parabola; UVL → cubic. A couple gives constant BM (rectangle).

23. (b) RA = wL/2 = 40 kN. SF at 2 m = 40 − 10 × 2 = 20 kN. (SF = 0 only at mid-span.)

D. Bending Stresses and Shear Stresses in Beams

24. (c) $\sigma/y = E/R \rightarrow \sigma = E\cdot y/R = 2\times10^5 \times 6/8000 = 150$ N/mm². y = half the thickness (2023 Q3 was this exact type).

25. (a) $Z = bd^2/6 = 100 \times 200^2/6 = 6.67 \times 10^5$ mm³. $\sigma = M/Z = 20 \times 10^6/6.67 \times 10^5 = 30$ N/mm².

26. (b) $M \propto Z$. $(b\cdot d^2/6) \div (d\cdot b^2/6) = d/b = 2$.

27. (a) Z(square)/Z(circle) ≈ 1.18 for equal area, so S1 is true. Bending stress is largest far from the NA, so material there is most effective, so S2 is true.

28. (c) Shear stress is MAXIMUM at the NA of a rectangular section (1.5 τavg). It is zero at the top and bottom edges.

29. (d) $\tau avg = V/A = 40{,}000/20{,}000 = 2$ N/mm². $\tau max = 1.5\ \tau avg = 3$ N/mm².

30. (d) All are true. The jump happens because $\tau = VA\bar{y}/(Ib)$ and the width b changes suddenly from flange width to web thickness.

E. Principal Stresses, Mohr's Circle and Theories of Failure

31. (a) Centre = (80+20)/2 = 50. Radius = $\sqrt{(30^2 + 40^2)} = 50$. $\sigma_{1,2} = 50 \pm 50 = 100, 0$. ($\tau max$ = radius = 50.)

32. (c) $\sigma n = \sigma \cos^2\theta = 100 \times (\sqrt{3}/2)^2 = 75$ N/mm². Shear on that plane = $(\sigma/2) \sin 2\theta = 43.3$ N/mm².

33. (a) In pure shear, Mohr's circle is centred at the origin with radius $\tau \rightarrow \sigma_1 = +\tau$, $\sigma_2 = -\tau$, at 45°. This is why brittle shafts break on 45° helical planes.

34. (b) Angles on Mohr's circle are TWICE the actual angles (2θ), so IV is false.

35. (b) Rankine–stress, Guest/Tresca–shear, St. Venant–strain, Haigh–total strain energy, Von Mises–distortion energy. (2023 asked 'Haigh's theory'.)

36. (b) $\tau max = (\sigma_1 - \sigma_3)/2 = (180 - (-30))/2 = 105$. Allowable = 200/2 = 100. 105 > 100 → failure. (2023 Q14 type — always use the largest minus the smallest, keeping the sign.)

37. (a) Both true, and R explains A. That is why Tresca/Von Mises are used for ductile materials and Rankine for brittle ones.

38. (b) $\varepsilon x = (\sigma x - \mu\sigma y)/E = (100 - 15)/2\times10^5 = 4.25 \times 10^{-4}$.

F. Deflection of Beams (asked heavily in 2023)

39. (d) $\delta = WL^3/3EI = 20 \times 8/(3 \times 4000) = 0.01333$ m = 13.33 mm.

40. (d) $\delta = 5wL^4/384EI \propto L^4 \rightarrow 2^4 = 16$ times. (For a central point load, $\delta \propto L^3 \rightarrow 8$ times.)

41. (a) Learn the deflection table in the box. A fixed beam deflects only ¼ as much as an SS beam under a central load (192 vs 48).

42. (b) In the conjugate beam: SF = slope of real beam; BM = deflection of real beam. So IV is false.

G. Columns and Struts

43. (a) Le changes from L to 2L. $P \propto 1/Le^2 \rightarrow 400/4 = 100$ kN.

44. (c) $k = \sqrt{(I/A)} = d/4 = 25$ mm. $\lambda = Le/k = 3000/25 = 120$.

45. (d) A column buckles about its weakest axis, so the LEAST moment of inertia is used (2023 Q7 also said 'least moment of inertia').

H. Torsion and Springs

46. (b) $P = 2\pi NT/60 = 2 \times 3.14 \times 300 \times 1000/60 = 31{,}400$ W = 31.4 kW.

47. (c) $Zp = J/R = (\pi d^4/32)/(d/2) = \pi d^3/16$, so IV is false ($\pi d^3/32$ is the Z for bending of a circle).

48. (d) Series: $1/k = 1/20 + 1/30 \rightarrow k = 12$ N/mm. $\delta = 120/12 = 10$ mm. (Parallel would give k = 50, δ = 2.4 mm — trap.)

I. Thin Cylinders and Lifting Machines

49. (c) $\sigma c = pd/2t = 2 \times 1000/20 = 100$ N/mm². Longitudinal = $pd/4t = 50$ N/mm².

50. (a) MA = 2000/125 = 16. $\eta = MA/VR = 16/40 = 40\%$. $\eta < 50\% \rightarrow$ self-locking (the load does not run back when the effort is removed).

Score analysis — fill this in and revise the weakest section first:

Section	Questions	Max	Correct	Wrong	Net (C − W/3)
A	1–10	10			
B	11–15	5			
C	16–23	8			
D	24–30	7			
E	31–38	8			
F	39–42	4			
G	43–45	3			
H	46–48	3			
I	49–50	2			
Total	1–50	**50**			

What your score means

Net 35 or more: ready — move to Module 2 (Fluid Mechanics).
Net 25 to 35: good — re-read the sections where you lost marks, solve 30 more MCQs on them, then move on.
Net below 25: spend 3–4 more days on Module 1 (formula boxes + Bansal solved examples), then retake this test after a week.
Write every wrong answer into the mistake notebook with the correct reason, and revise it on Sunday.